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Chapter 14

Linear Equations in One Variable

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Exercise 14(A)

Question 1(i)

For $\frac{x + 0.5}{6x - 1} = \frac{1}{2}$, the value of x is:

1. $\frac{1}{2}$

2. $-\frac{1}{2}$

3. 2



Chapter 19

Representing 3-D in 2-D

Chapter 20

Symmetry

4. -2

Answer

$$\frac{x + 0.5}{6x - 1} = \frac{1}{2}$$

On cross multiplying, we get

$$\Rightarrow (x + 0.5) \times 2 = 1 \times (6x - 1)$$

$$\Rightarrow 2 \times x + 2 \times 0.5 = 1 \times 6x - 1 \times 1$$

$$\Rightarrow 2x + 1 = 6x - 1$$

$$\Rightarrow 6x - 2x = 1 + 1$$

$$\Rightarrow 4x = 2$$

$$\Rightarrow x = \frac{2}{4}$$

$$\Rightarrow x = \frac{1}{2}$$

Hence, option 1 is the correct option.

Question 1(ii)

For $3(x + 1) - 4x = 0$, the value of x is:

1. $\frac{1}{3}$

2. $-\frac{1}{3}$

3. 3

4. -3

Answer

$$3(x + 1) - 4x = 0$$

$$\Rightarrow 3x + 3 - 4x = 0$$

$$\Rightarrow 3 - x = 0$$

$$\Rightarrow x = 3$$

Hence, option 3 is the correct option.

Question 1(iii)

For $7x - 8 = 5x + 2$, the value of x is:

1. $\frac{5}{6}$

2. $\frac{6}{5}$

3. 5

4. 8

Answer

$$7x - 8 = 5x + 2$$

$$\Rightarrow 7x - 5x = 2 + 8$$

$$\Rightarrow 2x = 10$$

$$\Rightarrow x = \frac{10}{2}$$

$$\Rightarrow x = 5$$

Hence, option 3 is the correct option.

Question 1(iv)

For $\frac{5}{x} = \frac{7}{x-4}$, the value of x is:

1. -10

2. 10

3. 5

4. -5

Answer

$$\frac{5}{x} = \frac{7}{x-4}$$

On cross multiplying, we get :

$$\Rightarrow 5 \times (x - 4) = 7 \times x$$

$$\Rightarrow 5 \times x - 5 \times 4 = 7x$$

$$\Rightarrow 5x - 20 = 7x$$

$$\Rightarrow 5x - 7x = 20$$

$$\Rightarrow -2x = 20$$

$$\Rightarrow x = -\frac{20}{2}$$

$$\Rightarrow x = -10$$

Hence, option 1 is the correct option.

Question 1(v)

If $\frac{x-2}{4} - \frac{x-3}{5} = 1$, the value of x is:

1. -42

2. 42

3. -18

4. 18

Answer

$$\frac{x-2}{4} - \frac{x-3}{5} = 1$$

Since L.C.M. of denominators 4 and 5 = 20, multiply each term by 20 to get:

$$\Rightarrow \frac{20(x-2)}{4} - \frac{20(x-3)}{5} = 1 \times 20$$

$$\Rightarrow 5(x-2) - 4(x-3) = 20$$

$$\Rightarrow 5x - 10 - 4x + 12 = 20$$

$$\Rightarrow x + 2 = 20$$

$$\Rightarrow x = 20 - 2$$

$$\Rightarrow x = 18$$

Hence, option 4 is the correct option.

Question 2

Solve for x :

$$\frac{3x + 2}{x - 6} = -7$$

Answer

$$\frac{3x + 2}{x - 6} = -7$$

On cross multiplying, we get :

$$\Rightarrow (3x + 2) = -7 \times (x - 6)$$

$$\Rightarrow 3x + 2 = -7x + 7 \times 6$$

$$\Rightarrow 3x + 2 = -7x + 42$$

$$\Rightarrow 3x + 7x = 42 - 2$$

$$\Rightarrow 10x = 40$$

$$\Rightarrow x = \frac{40}{10}$$

$$\Rightarrow x = 4$$

Hence, the value of x is 4.

Question 3

Solve for a :

$$3a - 4 = 2(4 - a)$$

Answer

$$3a - 4 = 2(4 - a)$$

$$\Rightarrow 3a - 4 = 8 - 2a$$

$$\Rightarrow 3a + 2a = 8 + 4$$

$$\Rightarrow 5a = 12$$

$$\Rightarrow a = \frac{12}{5}$$

$$\Rightarrow a = 2.4$$

Hence, the value of a is 2.4.

Question 4

Solve for b :

$$3(b - 4) = 2(4 - b)$$

Answer

$$3(b - 4) = 2(4 - b)$$

$$\Rightarrow 3b - 12 = 8 - 2b$$

$$\Rightarrow 3b + 2b = 8 + 12$$

$$\Rightarrow 5b = 20$$

$$\Rightarrow b = \frac{20}{5}$$

$$\Rightarrow b = 4$$

Hence, the value of b is 4.

Question 5

Solve for x :

$$\frac{x+2}{9} = \frac{x+4}{11}$$

Answer

$$\frac{x+2}{9} = \frac{x+4}{11}$$

On cross multiplying, we get :

$$\Rightarrow (x+2) \times 11 = (x+4) \times 9$$

$$\Rightarrow 11x + 22 = 9x + 36$$

$$\Rightarrow 11x - 9x = 36 - 22$$

$$\Rightarrow 2x = 14$$

$$\Rightarrow x = \frac{14}{2}$$

$$\Rightarrow x = 7$$

Hence, the value of x is 7.

Question 6

Solve for x :

$$\frac{x - 8}{5} = \frac{x - 12}{9}$$

Answer

$$\frac{x - 8}{5} = \frac{x - 12}{9}$$

On cross multiplying , we get :

$$\Rightarrow (x - 8) \times 9 = (x - 12) \times 5$$

$$\Rightarrow 9x - 72 = 5x - 60$$

$$\Rightarrow 9x - 5x = - 60 + 72$$

$$\Rightarrow 4x = 12$$

$$\Rightarrow x = \frac{12}{4}$$

$$\Rightarrow x = 3$$

Hence, the value of x is 3.

Question 7

Solve for x :

$$5(8x + 3) = 9(4x + 7)$$

Answer

$$5(8x + 3) = 9(4x + 7)$$

$$\Rightarrow 40x + 15 = 36x + 63$$

$$\Rightarrow 40x - 36x = 63 - 15$$

$$\Rightarrow 4x = 48$$

$$\Rightarrow x = \frac{48}{4}$$

$$\Rightarrow x = 12$$

Hence, the value of x is 12.

Question 8

Solve for x :

$$3(x + 1) = 12 + 4(x - 1)$$

Answer

$$3(x + 1) = 12 + 4(x - 1)$$

$$\Rightarrow 3x + 3 = 12 + 4x - 4$$

$$\Rightarrow 3x + 3 = 8 + 4x$$

$$\Rightarrow 3x - 4x = 8 - 3$$

$$\Rightarrow -x = 5$$

$$\Rightarrow x = -5$$

Hence, the value of x is - 5.

Question 9

Solve for x :

$$\frac{3x}{4} - \frac{1}{4}(x - 20) = \frac{x}{4} + 32$$

Answer

$$\frac{3x}{4} - \frac{1}{4}(x - 20) = \frac{x}{4} + 32$$

$$\Rightarrow \frac{3x}{4} - \frac{x}{4} + \frac{20}{4} = \frac{x}{4} + 32$$

$$\Rightarrow \frac{3x}{4} - \frac{x}{4} + 5 = \frac{x}{4} + 32$$

$$\Rightarrow \frac{3x - x}{4} + 5 = \frac{x}{4} + 32$$

$$\Rightarrow \frac{2x}{4} + 5 = \frac{x}{4} + 32$$

$$\Rightarrow \frac{2x}{4} - \frac{x}{4} = 32 - 5$$

$$\Rightarrow \frac{2x - x}{4} = 32 - 5$$

$$\Rightarrow \frac{x}{4} = 27$$

$$\Rightarrow x = 27 \times 4$$

$$\Rightarrow x = 108$$

Hence, the value of x is 108.

Question 10

Solve for a :

$$3a - \frac{1}{5} = \frac{a}{5} + 5\frac{2}{5}$$

Answer

$$3a - \frac{1}{5} = \frac{a}{5} + 5\frac{2}{5}$$

$$\Rightarrow 3a - \frac{1}{5} = \frac{a}{5} + \frac{27}{5}$$

$$\Rightarrow 3a - \frac{a}{5} = \frac{27}{5} + \frac{1}{5}$$

$$\Rightarrow \frac{15a}{5} - \frac{a}{5} = \frac{27}{5} + \frac{1}{5}$$

$$\Rightarrow \frac{15a - a}{5} = \frac{27 + 1}{5}$$

$$\Rightarrow \frac{14a}{5} = \frac{28}{5}$$

$$\Rightarrow a = \frac{28 \times 5}{5 \times 14}$$

$$\Rightarrow a = \frac{140}{70}$$

$$\Rightarrow a = 2$$

Hence, the value of a is 2.

Question 11

Solve for x :

$$\frac{x}{3} - 2\frac{1}{2} = \frac{4x}{9} - \frac{2x}{3}$$

Answer

$$\frac{x}{3} - 2\frac{1}{2} = \frac{4x}{9} - \frac{2x}{3}$$

$$\Rightarrow \frac{x}{3} - \frac{5}{2} = \frac{4x}{9} - \frac{2x}{3}$$

Since L.C.M. of denominators 3, 2 and 9 = 18, multiply each term by 18 to get:

$$\Rightarrow \frac{18 \times x}{3} - \frac{18 \times 5}{2} = \frac{18 \times 4x}{9} - \frac{18 \times 2x}{3}$$

$$\Rightarrow 6x - 9 \times 5 = 2 \times 4x - 6 \times 2x$$

$$\Rightarrow 6x - 45 = 8x - 12x$$

$$\Rightarrow 6x + 4x = 45$$

$$\Rightarrow 10x = 45$$

$$\Rightarrow x = \frac{45}{10}$$

$$\Rightarrow x = 4.5$$

Hence, the value of x is 4.5.

Question 12

Solve for y :

$$\frac{4(y + 2)}{5} = 7 + \frac{5y}{13}$$

Answer

$$\frac{4(y + 2)}{5} = 7 + \frac{5y}{13}$$

$$\Rightarrow \frac{4y + 8}{5} = 7 + \frac{5y}{13}$$

$$\Rightarrow \frac{4y}{5} + \frac{8}{5} = 7 + \frac{5y}{13}$$

$$\Rightarrow \frac{4y}{5} - \frac{5y}{13} = 7 - \frac{8}{5}$$

Since L.C.M. of denominators 5 and 13 = 65, multiply each term by 65 to get:

$$\Rightarrow \frac{4y \times 65}{5} - \frac{5y \times 65}{13} = 7 \times 65 - \frac{8 \times 65}{5}$$

$$\Rightarrow 4y \times 13 - 5y \times 5 = 455 - 8 \times 13$$

$$\Rightarrow 52y - 25y = 455 - 104$$

$$\Rightarrow 27y = 351$$

$$\Rightarrow y = \frac{351}{27}$$

$$\Rightarrow y = 13$$

Hence, the value of y is 13.

Question 13

Solve for a :

$$\frac{a+5}{6} - \frac{a+1}{9} = \frac{a+3}{4}$$

Answer

$$\frac{a+5}{6} - \frac{a+1}{9} = \frac{a+3}{4}$$

Since L.C.M. of denominators 6, 9 and 4 = 36, multiply each term by 36 to get:

$$\Rightarrow \frac{36(a+5)}{6} - \frac{36(a+1)}{9} = \frac{36(a+3)}{4}$$

$$\Rightarrow 6(a+5) - 4(a+1) = 9(a+3)$$

$$\Rightarrow (6a+30) - (4a+4) = (9a+27)$$

$$\Rightarrow 6a+30-4a-4 = 9a+27$$

$$\Rightarrow 2a+26 = 9a+27$$

$$\Rightarrow 2a-9a = 27-26$$

$$\Rightarrow -7a = 1$$

$$\Rightarrow a = -\frac{1}{7}$$

Hence, the value of a is $-\frac{1}{7}$.

Question 14

Solve for x :

$$\frac{2x-13}{5} - \frac{x-3}{11} = \frac{x-9}{5} + 1$$

Answer

$$\frac{2x - 13}{5} - \frac{x - 3}{11} = \frac{x - 9}{5} + 1$$

Since L.C.M. of denominators 5 and 11 = 55, multiply each term by 55 to get:

$$\Rightarrow \frac{(2x - 13) \times 55}{5} - \frac{(x - 3) \times 55}{11} = \frac{(x - 9) \times 55}{5} + 1 \times 55$$

$$\Rightarrow 11 (2x - 13) - 5 (x - 3) = 11 (x - 9) + 55$$

$$\Rightarrow (22x - 143) - (5x - 15) = (11x - 99) + 55$$

$$\Rightarrow 22x - 143 - 5x + 15 = 11x - 99 + 55$$

$$\Rightarrow 17x - 128 = 11x - 44$$

$$\Rightarrow 17x - 11x = 128 - 44$$

$$\Rightarrow 6x = 84$$

$$\Rightarrow x = \frac{84}{6}$$

$$\Rightarrow x = 14$$

Hence, the value of x is 14.

Question 15

Solve for x :

$$6(6x - 5) - 5(7x - 8) = 12(4 - x) + 1$$

Answer

$$6(6x - 5) - 5(7x - 8) = 12(4 - x) + 1$$

$$\Rightarrow 36x - 30 - 35x + 40 = 48 - 12x + 1$$

$$\Rightarrow 1x + 10 = 49 - 12x$$

$$\Rightarrow 1x + 12x = 49 - 10$$

$$\Rightarrow 13x = 39$$

$$\Rightarrow x = \frac{39}{13}$$

$$\Rightarrow x = 3$$

Hence, the value of x is 3.

Question 16

Solve for x :

$$(x - 5)(x + 3) = (x - 7)(x + 4)$$

Answer

$$(x - 5)(x + 3) = (x - 7)(x + 4)$$

$$\Rightarrow x(x + 3) - 5(x + 3) = x(x + 4) - 7(x + 4)$$

$$\Rightarrow x^2 + 3x - 5x - 15 = x^2 + 4x - 7x - 28$$

$$\Rightarrow x^2 - 2x - 15 = x^2 - 3x - 28$$

$$\Rightarrow x^2 - x^2 - 2x + 3x = -28 + 15$$

$$\Rightarrow 1x = -13$$

Hence, the value of x is - 13.

Question 17

Solve for x :

$$(x - 5)^2 - (x + 2)^2 = -2$$

Answer

$$(x - 5)^2 - (x + 2)^2 = -2$$

Using the formula,

$$[\because (x - y)^2 = x^2 + y^2 - 2xy]$$

And,

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$\Rightarrow x^2 + 5^2 - 2 \times x \times 5 - (x^2 + 2^2 + 2 \times x \times 2) = -2$$

$$\Rightarrow x^2 + 25 - 10x - (x^2 + 4 + 4x) = -2$$

$$\Rightarrow x^2 + 25 - 10x - x^2 - 4 - 4x = -2$$

$$\Rightarrow 21 - 14x = -2$$

$$\Rightarrow -14x = -2 - 21$$

$$\Rightarrow -14x = -23$$

$$\Rightarrow x = \frac{23}{14}$$

$$\Rightarrow x = 1\frac{9}{14}$$

Hence, the value of x is $1\frac{9}{14}$.

Question 18

Solve for x :

$$(x - 1)(x + 6) - (x - 2)(x - 3) = 3$$

Answer

$$(x - 1)(x + 6) - (x - 2)(x - 3) = 3$$

$$\Rightarrow x(x + 6) - 1(x + 6) - x(x - 3) + 2(x - 3) = 3$$

$$\Rightarrow x^2 + 6x - 1x - 6 - x^2 + 3x + 2x - 6 = 3$$

$$\Rightarrow x^2 - x^2 + 6x - 1x + 3x + 2x - 6 - 6 = 3$$

$$\Rightarrow 10x - 12 = 3$$

$$\Rightarrow 10x = 3 + 12$$

$$\Rightarrow 10x = 15$$

$$\Rightarrow x = \frac{15}{10}$$

$$\Rightarrow x = \frac{3}{2}$$

$$\Rightarrow x = 1\frac{1}{2}$$

Hence, the value of x is $1\frac{1}{2}$.

Question 19

Solve for x :

$$\frac{3x}{x+6} - \frac{x}{x+5} = 2$$

Answer

$$\frac{3x}{x+6} - \frac{x}{x+5} = 2$$

Since L.C.M. of denominators $(x+6)$ and $(x+5) = (x+6)(x+5)$, multiply each term with $(x+6)(x+5)$ to get:

$$\Rightarrow \frac{3x(x+5)(x+6)}{(x+6)} - \frac{x(x+5)(x+6)}{(x+5)} = 2(x+5)(x+6)$$

$$\Rightarrow 3x(x+5) - x(x+6) = 2(x+6)(x+5)$$

$$\Rightarrow (3x^2 + 15x) - (x^2 + 6x) = 2(x^2 + 6x + 5x + 30)$$

$$\Rightarrow 3x^2 + 15x - x^2 - 6x = 2(x^2 + 11x + 30)$$

$$\Rightarrow 2x^2 + 9x = 2x^2 + 22x + 60$$

$$\Rightarrow 2x^2 - 2x^2 + 9x - 22x = 60$$

$$\Rightarrow -13x = 60$$

$$\Rightarrow x = -\frac{60}{13}$$

$$\Rightarrow x = -4\frac{8}{13}$$

Hence, the value of x is $-4\frac{8}{13}$.

Question 20

Solve for x :

$$\frac{1}{x-1} + \frac{2}{x-2} = \frac{3}{x-3}$$

Answer

$$\frac{1}{x-1} + \frac{2}{x-2} = \frac{3}{x-3}$$

Since L.C.M. of denominators $(x-1)$, $(x-2)$ and $(x-3) = (x-1)(x-2)(x-3)$, multiply each term with $(x-1)(x-2)(x-3)$ to get :

$$\Rightarrow \frac{1(x-1)(x-2)(x-3)}{(x-1)} + \frac{2(x-1)(x-2)(x-3)}{(x-2)} = \frac{3(x-1)(x-2)(x-3)}{x-3}$$

$$\Rightarrow (x-2)(x-3) + 2(x-1)(x-3) = 3(x-1)(x-2)$$

$$\Rightarrow (x^2 - 2x - 3x + 6) + 2(x^2 - 1x - 3x + 3) = 3(x^2 - 1x - 2x + 2)$$

$$\Rightarrow (x^2 - 5x + 6) + 2(x^2 - 4x + 3) = 3(x^2 - 3x + 2)$$

$$\Rightarrow x^2 - 5x + 6 + 2x^2 - 8x + 6 = 3x^2 - 9x + 6$$

$$\Rightarrow 3x^2 - 13x + 12 = 3x^2 - 9x + 6$$

$$\Rightarrow 3x^2 - 3x^2 - 13x + 9x = 6 - 12$$

$$\Rightarrow -4x = -6$$

$$\Rightarrow x = \frac{6}{4}$$

$$\Rightarrow x = \frac{3}{2}$$

$$\Rightarrow x = 1\frac{1}{2}$$

Hence, the value of x is $1\frac{1}{2}$.

Question 21

Solve for x :

$$\frac{x-1}{7x-14} = \frac{x-3}{7x-26}$$

Answer

$$\frac{x-1}{7x-14} = \frac{x-3}{7x-26}$$

By cross multiplying, we get :

$$\Rightarrow (x-1)(7x-26) = (x-3)(7x-14)$$

$$\Rightarrow x(7x-26) - 1(7x-26) = x(7x-14) - 3(7x-14)$$

$$\Rightarrow 7x^2 - 26x - 7x + 26 = 7x^2 - 14x - 21x + 42$$

$$\Rightarrow 7x^2 - 33x + 26 = 7x^2 - 35x + 42$$

$$\Rightarrow 7x^2 - 7x^2 - 33x + 35x = 42 - 26$$

$$\Rightarrow 2x = 16$$

$$\Rightarrow x = \frac{16}{2}$$

$$\Rightarrow x = 8$$

Hence, the value of x is 8.

Question 22

Solve for x :

$$\frac{1}{x-1} - \frac{1}{x} = \frac{1}{x+3} - \frac{1}{x+4}$$

Answer

$$\frac{1}{x-1} - \frac{1}{x} = \frac{1}{x+3} - \frac{1}{x+4}$$

Since L.C.M. of denominators x and (x - 1) = x(x - 1) and L.C.M. of (x + 3) and (x + 4) = (x + 3)(x + 4),

$$\Rightarrow \frac{1 \times x}{x(x-1)} - \frac{1 \times (x-1)}{x(x-1)} = \frac{1 \times (x+4)}{(x+3) \times (x+4)} - \frac{1 \times (x+3)}{(x+4) \times (x+3)}$$

$$\Rightarrow \frac{x}{x^2-x} - \frac{x-1}{x^2-x} = \frac{x+4}{x^2+3x+4x+12} - \frac{x+3}{x^2+3x+4x+12}$$

$$\Rightarrow \frac{x - (x-1)}{x^2-x} = \frac{x+4}{x^2+7x+12} - \frac{x+3}{x^2+7x+12}$$

$$\Rightarrow \frac{x - x + 1}{x^2 - x} = \frac{x + 4}{x^2 + 7x + 12} - \frac{x + 3}{x^2 + 7x + 12}$$

$$\Rightarrow \frac{x - x + 1}{x^2 - x} = \frac{(x + 4) - (x + 3)}{x^2 + 7x + 12}$$

$$\Rightarrow \frac{1}{x^2 - x} = \frac{x + 4 - x - 3}{x^2 + 7x + 12}$$

$$\Rightarrow \frac{1}{x^2 - x} = \frac{1}{x^2 + 7x + 12}$$

By cross multiplying, we get :

$$\Rightarrow x^2 + 7x + 12 = x^2 - x$$

$$\Rightarrow x^2 - x^2 + 7x + x = -12$$

$$\Rightarrow 8x = -12$$

$$\Rightarrow x = -\frac{12}{8}$$

$$\Rightarrow x = -\frac{3}{2}$$

$$\Rightarrow x = -1\frac{1}{2}$$

Hence, the value of x is $-1\frac{1}{2}$.

Question 23

Solve :

$$\frac{2x}{3} - \frac{x-1}{6} + \frac{7x-1}{4} = 2\frac{1}{6}. \text{ Hence, find the value of 'a', if } \frac{1}{a} + 5x = 8.$$

Answer

$$\begin{aligned} \frac{2x}{3} - \frac{x-1}{6} + \frac{7x-1}{4} &= 2\frac{1}{6} \\ \Rightarrow \frac{2x}{3} - \frac{x-1}{6} + \frac{7x-1}{4} &= \frac{13}{6} \end{aligned}$$

Since L.C.M. of denominators 3, 6 and 4 = 12, multiply each term with 12 to get:

$$\Rightarrow \frac{2x \times 12}{3} - \frac{(x-1) \times 12}{6} + \frac{(7x-1) \times 12}{4} = \frac{13 \times 12}{6}$$

$$\Rightarrow 2x \times 4 - (x-1) \times 2 + (7x-1) \times 3 = 13 \times 2$$

$$\Rightarrow 8x - (2x - 2) + (21x - 3) = 26$$

$$\Rightarrow 8x - 2x + 2 + 21x - 3 = 26$$

$$\Rightarrow 27x - 1 = 26$$

$$\Rightarrow 27x = 26 + 1$$

$$\Rightarrow 27x = 27$$

$$\Rightarrow x = \frac{27}{27}$$

$$\Rightarrow x = 1$$

Now, when $x = 1$

$$\frac{1}{a} + 5x = 8$$

$$\Rightarrow \frac{1}{a} + 5 \times 1 = 8$$

$$\Rightarrow \frac{1}{a} + 5 = 8$$

$$\Rightarrow \frac{1}{a} = 8 - 5$$

$$\Rightarrow \frac{1}{a} = 3$$

$$\Rightarrow a = \frac{1}{3}$$

Hence, the value of x is 1 and a is $\frac{1}{3}$.

Question 24

Solve :

$$\frac{4-3x}{5} + \frac{7-x}{3} + 4\frac{1}{3} = 0. \text{ Hence, find the value of 'p', if } 3p - 2x + 1 = 0.$$

Answer

$$\frac{4-3x}{5} + \frac{7-x}{3} + 4\frac{1}{3} = 0$$

$$\Rightarrow \frac{4-3x}{5} + \frac{7-x}{3} + \frac{13}{3} = 0$$

Since L.C.M. of denominators 5 and 3 = 15, multiply each term with 15 to get:

$$\Rightarrow \frac{(4 - 3x) \times 15}{5} + \frac{(7 - x) \times 15}{3} + \frac{13 \times 15}{3} = 0 \times 15$$

$$\Rightarrow (4 - 3x) \times 3 + (7 - x) \times 5 + 13 \times 5 = 0$$

$$\Rightarrow (12 - 9x) + (35 - 5x) + 65 = 0$$

$$\Rightarrow 12 - 9x + 35 - 5x + 65 = 0$$

$$\Rightarrow 112 - 14x = 0$$

$$\Rightarrow 14x = 112$$

$$\Rightarrow x = \frac{112}{14}$$

$$\Rightarrow x = 8$$

Now, when $x = 8$

$$3p - 2x + 1 = 0$$

$$\Rightarrow 3p - 2 \times 8 + 1 = 0$$

$$\Rightarrow 3p - 16 + 1 = 0$$

$$\Rightarrow 3p - 15 = 0$$

$$\Rightarrow 3p = 15$$

$$\Rightarrow p = \frac{15}{3}$$

$$\Rightarrow p = 5$$

Hence, the value of x is 8 and p is 5.

Question 25

Solve :

$$0.25 + \frac{1.95}{x} = 0.9.$$

Answer

$$0.25 + \frac{1.95}{x} = 0.9$$

$$\Rightarrow 0.25 \times x + 1.95 = 0.9 \times x$$

$$\Rightarrow 0.25x + 1.95 = 0.9x$$

$$\Rightarrow 0.25x - 0.9x = -1.95$$

$$\Rightarrow -0.65x = -1.95$$

$$\Rightarrow x = \frac{1.95}{0.65}$$

$$\Rightarrow x = 3$$

Hence, the value of x is 3.**Question 26**

Solve :

$$5x - \left(4x + \frac{5x - 4}{7}\right) = \frac{4x - 14}{3}$$

Answer

$$\begin{aligned}5x - \left(4x + \frac{5x - 4}{7}\right) &= \frac{4x - 14}{3} \\ \Rightarrow 5x - 4x - \frac{5x - 4}{7} &= \frac{4x - 14}{3} \\ \Rightarrow x - \frac{5x - 4}{7} &= \frac{4x - 14}{3} \\ \Rightarrow \frac{x}{1} - \frac{5x - 4}{7} &= \frac{4x - 14}{3}\end{aligned}$$

Since L.C.M. of denominators 3 and 7 = 21, multiply each term with 21 to get:

$$\begin{aligned}\Rightarrow x \times 21 - \frac{(5x - 4) \times 21}{7} &= \frac{(4x - 14) \times 21}{3} \\ \Rightarrow 21x - (5x - 4) \times 3 &= (4x - 14) \times 7 \\ \Rightarrow 21x - (15x - 12) &= (28x - 98) \\ \Rightarrow 21x - 15x + 12 &= 28x - 98 \\ \Rightarrow 6x + 12 &= 28x - 98 \\ \Rightarrow 6x - 28x &= -98 - 12 \\ \Rightarrow -22x &= -110 \\ \Rightarrow x &= \frac{110}{22} \\ \Rightarrow x &= 5\end{aligned}$$

Hence, the value of x is 5.

Exercise 14(B)

Question 1(i)

The sum of three consecutive even numbers is 90; the middle number is:

1. 28
2. 30
3. 32
4. none of these

Answer

Let the 3 consecutive number be $(x - 2)$, x and $(x + 2)$.

$$(x - 2) + x + (x + 2) = 90$$

$$\Rightarrow x - 2 + x + x + 2 = 90$$

$$\Rightarrow 3x = 90$$

$$\Rightarrow x = \frac{90}{3}$$

$$\Rightarrow x = 30$$

Hence, option 2 is the correct option.

Question 1(ii)

The length of a rectangle is 8 m more than its breadth. If the perimeter of the rectangle is 40 m, the length of the rectangle is:

1. 6 m
2. 12 m
3. 14 m
4. 20 m

Answer

Let the breadth of the rectangle be a .

So, length of the rectangle = $8 + a$

Perimeter of the rectangle = 40

As we know, perimeter = $2(l + b)$

Putting the value,

$$\Rightarrow 40 = 2((8 + a) + a)$$

$$\Rightarrow 40 = 2(8 + a + a)$$

$$\Rightarrow 40 = 2(8 + 2a)$$

$$\Rightarrow 40 = 16 + 4a$$

$$\Rightarrow 4a = 40 - 16$$

$$\Rightarrow 4a = 24$$

$$\Rightarrow a = \frac{24}{4}$$

$$\Rightarrow a = 6$$

$$\text{Length} = a + 6 = 8 + 6 = 14 \text{ m}$$

Hence, option 3 is the correct option.

Question 1(iii)

If two whole numbers are in the ratio 5 : 3. If their difference is 18; the numbers are:

1. 27 and -45
2. 45 and -27
3. 45 and 27
4. -45 and -27

Answer

Given,

Two whole numbers are in the ratio 5 : 3.

Let the 2 numbers be $5x$ and $3x$.

Difference of two number be 18.

$$\Rightarrow 5x - 3x = 18$$

$$\Rightarrow 2x = 18$$

$$\Rightarrow x = \frac{18}{2}$$

$$\Rightarrow x = 9$$

So, the 2 numbers are $5 \times 9 = 45$ and $3 \times 9 = 27$.

Hence, option 3 is the correct option.

Question 1(iv)

Three more than twice a number is equal to four less than the number. The number is:

1. 7

2. -7

3. $\frac{1}{7}$

4. $-\frac{1}{7}$

Answer

Let the number be x .

So,

$$3 + 2x = x - 4$$

$$\Rightarrow 2x - x = -4 - 3$$

$$\Rightarrow x = -7$$

Hence, option 2 is the correct option.

Question 1(v)

The sum of two numbers is 70 and their difference is 16, the numbers are:

1. 43 and 27
2. 43 and -27
3. -43 and 27
4. -43 and -27

Answer

Let the two numbers be x and y.

So,

$$x + y = 70$$

$$x - y = 16$$

Adding two equations, we get

$$\Rightarrow (x + y) + (x - y) = 70 + 16$$

$$\Rightarrow x + y + x - y = 70 + 16$$

$$\Rightarrow 2x = 86$$

$$\Rightarrow x = \frac{86}{2}$$

$$\Rightarrow x = 43$$

So the other number be $y = 70 - x$

$$\Rightarrow y = 70 - 43$$

$$\Rightarrow y = 27$$

Hence, option 1 is the correct option.

Question 2

Fifteen less than 4 times a number is 9. Find the number.

Answer

Let the number be x .

According to the question,

$$\Rightarrow 4x - 15 = 9$$

$$\Rightarrow 4x = 9 + 15$$

$$\Rightarrow 4x = 24$$

$$\Rightarrow x = \frac{24}{4}$$

$$\Rightarrow x = 6$$

Hence, the number be 6.

Question 3

If Megha's age is increased by three times her age, the result is 60 years. Find her age.

Answer

Let Megha's age be x years.

So,

$$x + 3x = 60$$

$$\Rightarrow 4x = 60$$

$$\Rightarrow x = \frac{60}{4}$$

$$\Rightarrow x = 15$$

Hence, the age of Megha is 15 years.

Question 4

28 is 12 less than 4 times a number. Find the number.

Answer

Let the number be x .

So,

$$\Rightarrow 4x - 12 = 28$$

$$\Rightarrow 4x = 28 + 12$$

$$\Rightarrow 4x = 40$$

$$\Rightarrow x = \frac{40}{4}$$

$$\Rightarrow x = 10$$

Hence, the number be 10.

Question 5

Five less than 3 times a number is -20. Find the number.

Answer

Let the number be x.

So,

$$\Rightarrow 3x - 5 = -20$$

$$\Rightarrow 3x = -20 + 5$$

$$\Rightarrow 3x = -15$$

$$\Rightarrow x = -\frac{15}{3}$$

$$\Rightarrow x = -5$$

Hence, the number is - 5.

Question 6

Fifteen more than 3 times Neetu's age is the same as 4 times her age. How old is she ?

Answer

Let the Neetu's age be x years.

$$\Rightarrow 3x + 15 = 4x$$

$$\Rightarrow 4x - 3x = 15$$

$$\Rightarrow 1x = 15$$

Hence, the age of Neetu is 15 years.

Question 7

A number decreased by 30 is the same as 14 decreased by 3 times the number.

Find the number.

Answer

Let the number be x .

So,

$$\Rightarrow x - 30 = 14 - 3x$$

$$\Rightarrow x + 3x = 14 + 30$$

$$\Rightarrow 4x = 44$$

$$\Rightarrow x = \frac{44}{4}$$

$$\Rightarrow x = 11$$

Hence, the number is 11.

Question 8

A's salary is same as 4 times B's salary. If together they earn ₹ 3,750 a month, find the salary of each.

Answer

Let the B's salary be ₹ x .

So, A's salary = ₹ $4x$

Therefore,

A's salary + B's salary = 3,750

$$\Rightarrow 4x + x = 3,750$$

$$\Rightarrow 5x = 3,750$$

$$\Rightarrow x = \frac{3,750}{5}$$

$$\Rightarrow x = 750$$

So, A's salary = $4x$

$$= 4 \times 750$$

$$= 3,000$$

Hence, A's salary = ₹ 3,000 and B's salary = ₹ 750.

Question 9

Separate 178 into two parts so that the first part is 8 less than twice the second part.

Answer

Let the first part be x .

So the other part is $178 - x$.

$$\Rightarrow x = 2(178 - x) - 8$$

$$\Rightarrow x = 356 - 2x - 8$$

$$\Rightarrow x = 348 - 2x$$

$$\Rightarrow x + 2x = 348$$

$$\Rightarrow 3x = 348$$

$$\Rightarrow x = \frac{348}{3}$$

$$\Rightarrow x = 116$$

Other number = $178 - x$

$$= 178 - 116$$

$$= 62$$

Hence, the two numbers be 116 and 62.

Question 10

Six more than one-fourth of a number is two fifth of the number. Find the number.

Answer

Let the number be x .

So,

$$\Rightarrow 6 + \frac{1}{4}x = \frac{2}{5}x$$

$$\Rightarrow 6 = \frac{2}{5}x - \frac{1}{4}x$$

Since L.C.M. of denominators 5 and 4 = 20, multiply each term with 20 to get:

$$\Rightarrow 6 \times 20 = \frac{2 \times 20}{5}x - \frac{1 \times 20}{4}x$$

$$\Rightarrow 120 = 2 \times 4x - 1 \times 5x$$

$$\Rightarrow 120 = 8x - 5x$$

$$\Rightarrow 120 = 3x$$

$$\Rightarrow x = \frac{120}{3}$$

$$\Rightarrow x = 40$$

Hence, the number is 40.

Question 11

The length of a rectangle is twice its width. If its perimeter is 54 cm, find its length.

Answer

Let the breadth of the rectangle be b cm.

The length of a rectangle is twice its width.

$$l = 2b$$

Perimeter of the rectangle = 54cm

$$\text{Perimeter} = 2(l + b)$$

$$\Rightarrow 2(2b + b) = 54 \text{ cm}$$

$$\Rightarrow 2(3b) = 54 \text{ cm}$$

$$\Rightarrow 6b = 54 \text{ cm}$$

$$\Rightarrow b = \frac{54}{6} \text{ cm}$$

$$\Rightarrow b = 9 \text{ cm}$$

$$l = 2b$$

$$= 2 \times 9$$

$$= 18 \text{ cm}$$

Hence, length of the rectangle is 18cm.

Question 12

A rectangle's length is 5 cm less than twice its width. If the length is decreased by 5 cm and width is increased by 2 cm, the perimeter of the resulting rectangle will be 74 cm. Find the length and the width of the original rectangle.

Answer

Let the width of the rectangle be w cm

Length of the rectangle is 5 cm less than twice its width.

$$l = 2w - 5 \text{ cm}$$

$$\text{New width} = \text{old width} + 2$$

$$= w + 2 \text{ cm}$$

$$\text{New length} = \text{old length} - 5$$

$$= (2w - 5) - 5 \text{ cm}$$

$$= 2w - 5 - 5 \text{ cm}$$

$$= 2w - 10 \text{ cm}$$

$$\text{New perimeter} = 2(\text{new length} + \text{new width})$$

$$\Rightarrow 2((2w - 10) + (w + 2)) = 74 \text{ cm}$$

$$\Rightarrow 2(2w - 10 + w + 2) = 74 \text{ cm}$$

$$\Rightarrow 2(3w - 8) = 74 \text{ cm}$$

$$\Rightarrow 6w - 16 = 74 \text{ cm}$$

$$\Rightarrow 6w = 74 + 16 \text{ cm}$$

$$\Rightarrow 6w = 90 \text{ cm}$$

$$\Rightarrow w = \frac{90}{6} \text{ cm}$$

$$\Rightarrow w = 15 \text{ cm}$$

$$\text{Length} = 2w - 5 \text{ cm}$$

$$= 2 \times 15 - 5 \text{ cm}$$

$$= 30 - 5 \text{ cm}$$

$$= 25 \text{ cm}$$

Hence, the length is 25 cm and breadth is 15 cm.

Question 13

The sum of three consecutive odd numbers is 57. Find the numbers.

Answer

Let the three consecutive odd numbers be $(x - 2)$, x , $(x + 2)$.

So,

$$(x - 2) + x + (x + 2) = 57$$

$$\Rightarrow x - 2 + x + x + 2 = 57$$

$$\Rightarrow 3x = 57$$

$$\Rightarrow x = \frac{57}{3}$$

$$\Rightarrow x = 19$$

Other numbers = $(x - 2)$, $(x + 2)$

$$= (19 - 2), (19 + 2)$$

$$= 17, 21$$

Hence, the three numbers are 17, 19 and 21.

Question 14

A man's age is three times that of his son and in twelve years he will be twice as old as his son would be. What are their present ages?

Answer

Let the age of son be x years.

The man's age is three times that of his son, so it equals $3x$.

In twelve years, he will be twice as old as his son will be,

$$\text{Man's age} + 12 = 2(\text{son's age} + 12)$$

$$\Rightarrow 3x + 12 = 2(x + 12)$$

$$\Rightarrow 3x + 12 = 2x + 24$$

$$\Rightarrow 3x - 2x = 24 - 12$$

$$\Rightarrow x = 12$$

$$\text{Man's age} = 3x$$

$$= 3 \times 12$$

$$= 36$$

Hence, the man's age is 36 years and the son's age is 12 years.

Question 15

A man is 42 years old and his son is 12 years old. In how many years will the age of the son be half the age of the man at that time?

Answer

The age of man = 42 years

The age of son = 12 years

Let the son's age be half of the man's age after x years.

$$\text{Son's age} + x = \frac{1}{2} (\text{Man's age} + x)$$

$$\Rightarrow 12 + x = \frac{1}{2} (42 + x)$$

$$\Rightarrow 2 \times (12 + x) = 1 \times (42 + x)$$

$$\Rightarrow 24 + 2x = 42 + x$$

$$\Rightarrow 2x - x = 42 - 24$$

$$\Rightarrow x = 18$$

Hence, after 18 years, the son's age will be half of the man's age.

Question 16

A man completed a trip of 136 km in 8 hours. Some parts of the trip was covered at 15 km/hr and the remaining at 18 km/hr. Find the part of the trip covered at 18 km/hr.

Answer

Total distance = 136 km

Total time = 8 hours

Let the part of the trip that was covered at 15 km/h be x km.

So, the remaining part, $(136 - x)$ km, was covered at 18 km/hr.

$$\text{As we know, Time} = \frac{\text{Distance}}{\text{Speed}}$$

And

Total time = time taken to cover x km at 15 km/h + time taken to cover $(136 - x)$ km at 18 km/h.

$$\Rightarrow \frac{x}{15} + \frac{136 - x}{18} = 8$$

Since L.C.M. of 15 and 18 = 90, multiply each term with 90 to get:

$$\Rightarrow \frac{x \times 90}{15} + \frac{(136 - x) \times 90}{18} = 8 \times 90$$

$$\Rightarrow x \times 6 + (136 - x) \times 5 = 720$$

$$\Rightarrow 6x + 680 - 5x = 720$$

$$\Rightarrow x + 680 = 720$$

$$\Rightarrow x = 720 - 680$$

$$\Rightarrow x = 40$$

Other part = $136 - x$ km

$$= 136 - 40 \text{ km}$$

$$= 96 \text{ km}$$

Hence, the man covered 96 km at the speed of 18 km/hr.

Question 17

The difference of two numbers is 3 and the difference of their squares is 69.
Find the numbers.

Answer

Let the two numbers be x and y .

So,

$$x - y = 3 \dots\dots\dots(1)$$

And,

$$x^2 - y^2 = 69$$

Using the formula

$$[\because (x^2 - y^2) = (x - y)(x + y)]$$

$$\Rightarrow (x - y)(x + y) = 69$$

Using eq (1), we get

$$\Rightarrow 3(x + y) = 69$$

$$\Rightarrow x + y = \frac{69}{3}$$

$$\Rightarrow x + y = 23 \dots\dots\dots(2)$$

Adding (1) and (2)

$$\Rightarrow (x - y) + (x + y) = 3 + 23$$

$$\Rightarrow x - y + x + y = 26$$

$$\Rightarrow 2x = 26$$

$$\Rightarrow x = \frac{26}{2}$$

$$\Rightarrow x = 13$$

Using eq (2)

$$\Rightarrow y = 23 - 13$$

$$\Rightarrow y = 10$$

Hence, the numbers are 13 and 10.

Question 18

Two consecutive natural numbers are such that one-fourth of the smaller exceeds one-fifth of the greater by 1. Find the numbers.

Answer

Let the consecutive natural numbers be x and $(x + 1)$.

So,

$$\frac{1}{4} \times x - \frac{1}{5} \times (x + 1) = 1$$

$$\Rightarrow \frac{x}{4} - \frac{x + 1}{5} = 1$$

Since L.C.M. of denominators 5 and 4 = 20, multiply each term with 20 to get:

$$\Rightarrow \frac{x \times 20}{4} - \frac{(x + 1) \times 20}{5} = 1 \times 20$$

$$\Rightarrow x \times 5 - (x + 1) \times 4 = 20$$

$$\Rightarrow 5x - (4x + 4) = 20$$

$$\Rightarrow 5x - 4x - 4 = 20$$

$$\Rightarrow x - 4 = 20$$

$$\Rightarrow x = 20 + 4$$

$$\Rightarrow x = 24$$

$$\text{Other number} = (x + 1)$$

$$= 24 + 1$$

$$= 25$$

Hence, the numbers are 24 and 25.

Question 19

Three consecutive whole numbers are such that if they be divided by 5, 3 and 4 respectively, the sum of the quotients is 40. Find the numbers.

Answer

Let the three consecutive whole numbers be $(x - 1)$, x , $(x + 1)$.

According to the question, the sum of the quotients is 40.

$$\Rightarrow \frac{x - 1}{5} + \frac{x}{3} + \frac{x + 1}{4} = 40$$

Since L.C.M. of denominators 5, 3 and 4 = 60, multiply each term with 60 to get:

$$\Rightarrow \frac{(x-1) \times 60}{5} + \frac{x \times 60}{3} + \frac{(x+1) \times 60}{4} = 40 \times 60$$

$$\Rightarrow (x-1) \times 12 + x \times 20 + (x+1) \times 15 = 2400$$

$$\Rightarrow (12x - 12) + 20x + (15x + 15) = 2400$$

$$\Rightarrow 12x - 12 + 20x + 15x + 15 = 40$$

$$\Rightarrow 47x + 3 = 2400$$

$$\Rightarrow 47x = 2400 - 3$$

$$\Rightarrow 47x = 2397$$

$$\Rightarrow x = \frac{2397}{47}$$

$$\Rightarrow x = 51$$

Other number = $(x - 1)$, $(x + 1)$

$$= (51 - 1), (51 + 1)$$

$$= 50, 52$$

Hence, the numbers are 50, 51 and 52.

Question 20

If the same number be added to the numbers 5, 11, 15 and 31, the resulting numbers are in proportion. Find the number.

Answer

Let x be added to each number. The numbers will then be $5 + x$, $11 + x$, $15 + x$ and $31 + x$.

According to question,

$$\frac{5 + x}{11 + x} = \frac{15 + x}{31 + x}$$

By cross multiplying, we get :

$$\Rightarrow (5 + x)(31 + x) = (15 + x)(11 + x)$$

$$\Rightarrow 155 + 5x + 31x + x^2 = 165 + 11x + 15x + x^2$$

$$\Rightarrow 155 + 36x + x^2 = 165 + 26x + x^2$$

$$\Rightarrow 36x - 26x + x^2 - x^2 = 165 - 155$$

$$\Rightarrow 10x = 10$$

$$\Rightarrow x = \frac{10}{10}$$

$$\Rightarrow x = 1$$

Hence, 1 is the number that should be added to the given numbers.

Question 21

The present age of a man is twice that of his son. Eight years hence, their ages will be in the ratio 7 : 4. Find their present ages.

Answer

Let the age of son be x years.

The present age of a man is twice that of his son.

Man's age = $2x$

After eight years, their ages will be in the ratio 7 : 4.

$$\Rightarrow \frac{\text{Man's age after 8 years}}{\text{Son's age after 8 years}} = \frac{7}{4}$$

$$\Rightarrow \frac{\text{Man's present age} + 8 \text{ years}}{\text{Son's present age} + 8 \text{ years}} = \frac{7}{4}$$

$$\Rightarrow \frac{2x + 8}{x + 8} = \frac{7}{4}$$

By cross multiplying, we get

$$\Rightarrow 4(2x + 8) = 7(x + 8)$$

$$\Rightarrow 8x + 32 = 7x + 56$$

$$\Rightarrow 8x - 7x = 56 - 32$$

$$\Rightarrow x = 24$$

Present son's age = 24 years

Present man's age = $2x$ years

$$= 2 \times 24 \text{ years}$$

$$= 48 \text{ years}$$

Hence, son's age is 24 years and man's age is 48 years.

Test Yourself

Question 1(i)

For $\frac{x}{2} + \frac{x}{3} = 15$; the value of x is:

1. 18
2. 90
3. 15
4. 45

Answer

$$\frac{x}{2} + \frac{x}{3} = 15$$

Since L.C.M. of denominators 2 and 3 = 6, multiply each term with 6 to get:

$$\Rightarrow \frac{x \times 6}{2} + \frac{x \times 6}{3} = 15 \times 6$$

$$\Rightarrow x \times 3 + x \times 2 = 90$$

$$\Rightarrow 3x + 2x = 90$$

$$\Rightarrow 5x = 90$$

$$\Rightarrow x = \frac{90}{5}$$

$$\Rightarrow x = 18$$

Hence, option 1 is the correct option.

Question 1(ii)

The sum of three consecutive odd whole numbers is 87. The largest of these is:

1. 27

2. 31

3. 29

4. 25

Answer

Let the three consecutive odd whole numbers be $(x - 2)$, x and $(x + 2)$.

According to question,

$$\Rightarrow (x - 2) + x + (x + 2) = 87$$

$$\Rightarrow x - 2 + x + x + 2 = 87$$

$$\Rightarrow 3x = 87$$

$$\Rightarrow x = \frac{87}{3}$$

$$\Rightarrow x = 29$$

Other numbers are $(x - 2)$, $(x + 2)$

$$= (29 - 2), (29 + 2)$$

$$= 27, 31$$

The larger number is 31.

Hence, option 2 is the correct option.

Question 1(iii)

If $\frac{x + 3}{(x + 4) + 3} = \frac{4}{5}$, the value of x is:

1. 12

2. -13

3. -12

4. 13

Answer

$$\frac{x + 3}{(x + 4) + 3} = \frac{4}{5}$$

$$\Rightarrow \frac{x + 3}{x + 4 + 3} = \frac{4}{5}$$

$$\Rightarrow \frac{x+3}{x+7} = \frac{4}{5}$$

By cross multiplying, we get :

$$\Rightarrow 5(x+3) = 4(x+7)$$

$$\Rightarrow 5x + 15 = 4x + 28$$

$$\Rightarrow 5x - 4x = 28 - 15$$

$$\Rightarrow x = 13$$

Hence, option 4 is the correct option.

Question 1(iv)

When four consecutive integers are added, their sum is -46. The smallest integer is:

1. 13

2. -13

3. 10

4. -10

Answer

Let the four consecutive integers be $(x - 1)$, x , $(x + 1)$ and $(x + 2)$.

$$\Rightarrow (x - 1) + x + (x + 1) + (x + 2) = -46$$

$$\Rightarrow x - 1 + x + x + 1 + x + 2 = -46$$

$$\Rightarrow 4x + 2 = -46$$

$$\Rightarrow 4x = -46 - 2$$

$$\Rightarrow 4x = -48$$

$$\Rightarrow x = -\frac{48}{4}$$

$$\Rightarrow x = -12$$

Other numbers are $(x - 1)$, $(x + 1)$, $(x + 2)$

$$= (-12 - 1), (-12 + 1), (-12 + 2)$$

$$= -13, -11, -10$$

Hence, option 2 is the correct option.

Question 1(v)

The perimeter of a rectangular plot is 560 m and the length of the plot is three times its width; the length of the plot is:

1. 140 m

2. 210 m

3. 280 m

4. $186\frac{2}{3}$ m

Answer

Let the width of the rectangular plot be w .

Length of the rectangular plot = $3w$

Perimeter = 560 m

As we know, perimeter of a rectangle = $2(l + w)$

$$\Rightarrow 2(3w + w) = 560 \text{ m}$$

$$\Rightarrow 2 \times 4w = 560 \text{ m}$$

$$\Rightarrow 8w = 560 \text{ m}$$

$$\Rightarrow w = \frac{560}{8} \text{ m}$$

$$\Rightarrow w = 70 \text{ m}$$

Length of rectangle = $3w$

$$= 3 \times 70 \text{ m}$$

$$= 210 \text{ m}$$

Hence, option 2 is the correct option.

Question 1(vi)

Statement 1: $\frac{ax + b}{cx + d} = \frac{p}{q}$.

$$\Rightarrow q(cx + d) = p(ax + b)$$

This process is cross-multiplication.

Statement 2: $ax + b = c$ becomes $ax = c - b$ after transposition.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

Given, $\frac{ax + b}{cx + d} = \frac{p}{q}$

On cross-multiplication,

$$\Rightarrow q(ax + b) = p(cx + d)$$

So, statement 1 is false.

Transposition is a fundamental algebraic technique used to isolate a variable by moving terms from one side of an equation to the other. When a term is transposed, its sign changes;

- A term added to one side gets subtracted on the other.
- A term subtracted from one side gets added on the other.

$\therefore ax + b = c$ becomes $ax = c - b$ after transposition.

So, statement 2 is true.

$\therefore$ Statement 1 is false, and statement 2 is true.

Hence, option 4 is the correct option.

Question 1(vii)

Assertion (A) : $(2x + 5)^2 - (2x - 5)^2 = 40$ is a linear equation in terms of one variable.

Reason (R) : An equation in which the greatest exponent of the variable after simplification is one is called a linear equation.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

An equation in which the greatest exponent of the variable after simplification is one is called a linear equation.

This definition aligns with the standard definition of a linear equation. A linear equation in one variable is of the form $ax + b = 0$, where $a \neq 0$ and the highest power of x is 1.

So, reason (R) is true.

Given,

$$\Rightarrow (2x + 5)^2 - (2x - 5)^2 = 40$$

$$\Rightarrow 4x^2 + 25 + 20x - (4x^2 + 25 - 20x) = 40$$

$$\Rightarrow 4x^2 + 25 + 20x - 4x^2 - 25 + 20x = 40$$

$$\Rightarrow 40x = 40$$

$$\Rightarrow 40x - 40 = 0$$

The above equation is in the form $ax + b = 0$, where $a = 40$, $b = -40$ and the highest power of x is 1.

So, assertion (A) is true.

$\therefore$ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 1(viii)

Assertion (A) : The solution of : $\frac{x}{2} + \frac{x}{3} = 15$ is 18.

Reason (R) : In the process of solving a linear equation, only the method of transposition is applied.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

$$\Rightarrow \frac{x}{2} + \frac{x}{3} = 15$$

$$\Rightarrow \frac{3x + 2x}{6} = 15$$

$$\Rightarrow \frac{5x}{6} = 15$$

$$\Rightarrow x = \frac{6 \times 15}{5}$$

$$\Rightarrow x = \frac{90}{5}$$

$$\Rightarrow x = 18$$

So, assertion (A) is true.

In the process of solving a linear equation, multiple methods can be applied, not just transposition.

So, reason (R) is false.

$\therefore$ A is true, but R is false.

Hence, option 3 is the correct option.

Question 1(ix)

Assertion (A) : The solution of : $\frac{x-3}{x+4} = \frac{x+1}{x-2}$ is 5.

Reason (R) : $\frac{ax+b}{cx+d} = \frac{p}{q}$.

$$\Rightarrow q(ax + b) = p(cx + d)$$

This process is cross-multiplication.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given, $\frac{ax + b}{cx + d} = \frac{p}{q}$

On cross-multiplication,

$$\Rightarrow q(ax + b) = p(cx + d)$$

So, reason (R) is true.

Given,

$$\Rightarrow \frac{x - 3}{x + 4} = \frac{x + 1}{x - 2}$$

$$\Rightarrow (x - 3) \times (x - 2) = (x + 4) \times (x + 1)$$

$$\Rightarrow x(x - 2) - 3(x - 2) = x(x + 1) + 4(x + 1)$$

$$\Rightarrow x^2 - 2x - 3x + 6 = x^2 + x + 4x + 4$$

$$\Rightarrow x^2 - 5x + 6 = x^2 + 5x + 4$$

$$\Rightarrow x^2 - 5x + 6 - x^2 - 5x - 4 = 0$$

$$\Rightarrow -10x + 2 = 0$$

$$\Rightarrow 10x = 2$$

$$\Rightarrow x = \frac{2}{10} = \frac{1}{5}.$$

So, assertion (A) is false.

$\therefore$ A is false, but R is true.

Hence, option 4 is the correct option.

Question 1(x)

Assertion (A) : $7x = -21$

$$\Rightarrow \frac{7x}{7} = -\frac{21}{7} \Rightarrow x = -3$$

Reason (R) : An equation remains unchanged on dividing both sides by the same (non-zero) number.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

In algebra, on dividing both sides of an equation by the same non-zero number, the equation remains the same.

So, reason (R) is true.

Given, $7x = -21$

Dividing both sides by 7, we get

$$\Rightarrow \frac{7x}{7} = -\frac{21}{7}$$

$$\Rightarrow x = -3$$

So, assertion (A) is true.

$\therefore$ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 2(i)

Solve:

$$\frac{1}{3}x - 6 = \frac{5}{2}$$

Answer

$$\frac{1}{3}x - 6 = \frac{5}{2}$$

$$\Rightarrow \frac{1}{3}x = \frac{5}{2} + 6$$

$$\Rightarrow \frac{1}{3}x = \frac{5}{2} + \frac{6}{1}$$

Since L.C.M. of denominators 3, 2 and 1 = 6, multiply each term with 6 to get:

$$\Rightarrow \frac{1 \times 6}{3}x = \frac{5 \times 6}{2} + \frac{6 \times 6}{1}$$

$$\Rightarrow 1 \times 2x = 5 \times 3 + 6 \times 6$$

$$\Rightarrow 2x = 15 + 36$$

$$\Rightarrow 2x = 51$$

$$\Rightarrow x = \frac{51}{2}$$

$$\Rightarrow x = 25\frac{1}{2}$$

Hence, the value of x is $25\frac{1}{2}$.

Question 2(ii)

Solve:

$$\frac{2x}{3} - \frac{3x}{8} = \frac{7}{12}$$

Answer

$$\frac{2x}{3} - \frac{3x}{8} = \frac{7}{12}$$

Since L.C.M. of denominators 3 and 8 = 24, multiply each term with 24 to get:

$$\Rightarrow \frac{2x \times 24}{3} - \frac{3x \times 24}{8} = \frac{7 \times 24}{12}$$

$$\Rightarrow 2x \times 8 - 3x \times 3 = 7 \times 2$$

$$\Rightarrow 16x - 9x = 14$$

$$\Rightarrow 7x = 14$$

$$\Rightarrow x = \frac{14}{7}$$

$$\Rightarrow x = 2$$

Hence, the value of x is 2.

Question 2(iii)

Solve:

$$(x + 2)(x + 3) + (x - 3)(x - 2) - 2x(x + 1) = 0$$

Answer

$$(x + 2)(x + 3) + (x - 3)(x - 2) - 2x(x + 1) = 0$$

$$\Rightarrow x(x + 3) + 2(x + 3) + x(x - 2) - 3(x - 2) - 2x(x + 1) = 0$$

$$\Rightarrow x^2 + 3x + 2x + 6 + x^2 - 2x - 3x + 6 - 2x^2 - 2x = 0$$

$$\Rightarrow (x^2 + x^2 - 2x^2) + (3x + 2x - 2x - 3x - 2x) + (6 + 6) = 0$$

$$\Rightarrow -2x + 12 = 0$$

$$\Rightarrow 2x = 12$$

$$\Rightarrow x = \frac{12}{2}$$

$$\Rightarrow x = 6$$

Hence, the value of x is 6.

Question 2(iv)

Solve:

$$\frac{1}{10} - \frac{7}{x} = 35$$

Answer

$$\frac{1}{10} - \frac{7}{x} = 35$$

$$\Rightarrow \frac{7}{x} = \frac{1}{10} - 35$$

$$\Rightarrow \frac{7}{x} = \frac{1}{10} - \frac{35}{1}$$

Since L.C.M. of denominators 10 and 1 = 10, multiply each term with 10 to get:

$$\Rightarrow \frac{7 \times 10}{x} = \frac{1 \times 10}{10} - \frac{35 \times 10}{1}$$

$$\Rightarrow \frac{70}{x} = 1 - 350$$

$$\Rightarrow \frac{70}{x} = -349$$

$$\Rightarrow x = -\frac{70}{349}$$

Hence, the value of x is $-\frac{70}{349}$.**Question 2(v)**

Solve:

$$13(x - 4) - 3(x - 9) - 5(x + 4) = 0$$

Answer

$$13(x - 4) - 3(x - 9) - 5(x + 4) = 0$$

$$\Rightarrow 13x - 52 - 3x + 27 - 5x - 20 = 0$$

$$\Rightarrow (13x - 3x - 5x) + (-52 + 27 - 20) = 0$$

$$\Rightarrow 5x - 45 = 0$$

$$\Rightarrow 5x = 45$$

$$\Rightarrow x = \frac{45}{5}$$

$$\Rightarrow x = 9$$

Hence, the value of x is 9.

Question 2(vi)

Solve:

$$x + 7 - \frac{8x}{3} = \frac{17x}{6} - \frac{5x}{8}$$

Answer

$$x + 7 - \frac{8x}{3} = \frac{17x}{6} - \frac{5x}{8}$$

Since L.C.M. of denominators 6 and 8 = 24, multiply each term with 24 to get:

$$\Rightarrow x \times 24 + 7 \times 24 - \frac{8x \times 24}{3} = \frac{17x \times 24}{6} - \frac{5x \times 24}{8}$$

$$\Rightarrow 24x + 168 - 8x \times 8 = 17x \times 4 - 5x \times 3$$

$$\Rightarrow 24x + 168 - 64x = 68x - 15x$$

$$\Rightarrow 168 - 40x = 53x$$

$$\Rightarrow -53x - 40x = -168$$

$$\Rightarrow -93x = -168$$

$$\Rightarrow x = \frac{168}{93}$$

$$\Rightarrow x = \frac{56}{31}$$

$$\Rightarrow x = 1\frac{25}{31}$$

Hence, the value of x is $1\frac{25}{31}$

Question 2(vii)

Solve:

$$\frac{3x - 2}{4} - \frac{2x + 3}{3} = \frac{2}{3} - x$$

Answer

$$\frac{3x - 2}{4} - \frac{2x + 3}{3} = \frac{2}{3} - x$$

$$\Rightarrow \frac{3x - 2}{4} - \frac{2x + 3}{3} + x = \frac{2}{3}$$

$$\Rightarrow \frac{3x - 2}{4} - \frac{2x + 3}{3} + \frac{x}{1} = \frac{2}{3}$$

Since L.C.M. of denominators 3 and 4 = 12, multiply each term with 12 to get:

$$\Rightarrow \frac{12(3x - 2)}{4} - \frac{12(2x + 3)}{3} + \frac{x \times 12}{1} = \frac{2 \times 12}{3}$$

$$\Rightarrow 3(3x - 2) - 4(2x + 3) + x \times 12 = 2 \times 4$$

$$\Rightarrow (9x - 6) - (8x + 12) + 12x = 8$$

$$\Rightarrow 9x - 6 - 8x - 12 + 12x = 8$$

$$\Rightarrow 13x - 18 = 8$$

$$\Rightarrow 13x = 8 + 18$$

$$\Rightarrow 13x = 26$$

$$\Rightarrow x = \frac{26}{13}$$

$$\Rightarrow x = 2$$

Hence, the value of x is 2.

Question 2(viii)

Solve:

$$\frac{x+2}{6} - \left(\frac{11-x}{3} - \frac{1}{4} \right) = \frac{3x-4}{12}$$

Answer

$$\frac{x+2}{6} - \left(\frac{11-x}{3} - \frac{1}{4} \right) = \frac{3x-4}{12}$$

$$\Rightarrow \frac{x+2}{6} - \frac{11-x}{3} + \frac{1}{4} = \frac{3x-4}{12}$$

Since L.C.M. of denominators 6, 3 and 4 = 12, multiply each term with 12 to get:

$$\Rightarrow \frac{12(x+2)}{6} - \frac{12(11-x)}{3} + \frac{1 \times 12}{4} = \frac{12(3x-4)}{12}$$

$$\Rightarrow 2(x+2) - 4(11-x) + 1 \times 3 = 1(3x-4)$$

$$\Rightarrow (2x+4) - (44-4x) + 3 = 3x-4$$

$$\Rightarrow 2x+4-44+4x+3=3x-4$$

$$\Rightarrow 6x-37=3x-4$$

$$\Rightarrow 6x-3x=37-4$$

$$\Rightarrow 3x=33$$

$$\Rightarrow x = \frac{33}{3}$$

$$\Rightarrow x = 11$$

Hence, the value of x is 11.

Question 2(ix)

Solve:

$$\frac{2}{5x} - \frac{5}{3x} = \frac{1}{15}$$

Answer

$$\frac{2}{5x} - \frac{5}{3x} = \frac{1}{15}$$

Since L.C.M. of denominators $5x$ and $3x = 15x$, multiply each term with $15x$ to get:

$$\Rightarrow \frac{2 \times 15x}{5x} - \frac{5 \times 15x}{3x} = \frac{1 \times 15x}{15}$$

$$\Rightarrow 2 \times 3 - 5 \times 5 = 1 \times x$$

$$\Rightarrow 6 - 25 = x$$

$$\Rightarrow -19 = x$$

Hence, the value of x is -19 .

Question 2(x)

Solve:

$$\frac{x+2}{3} - \frac{x+1}{5} = \frac{x-3}{4} - 1$$

Answer

$$\frac{x+2}{3} - \frac{x+1}{5} = \frac{x-3}{4} - 1$$

$$\Rightarrow \frac{x+2}{3} - \frac{x+1}{5} - \frac{x-3}{4} = -1$$

Since L.C.M. of denominators 3 , 5 and $4 = 60$, multiply each term with 60 to get:

$$\Rightarrow \frac{60(x+2)}{3} - \frac{60(x+1)}{5} - \frac{60(x-3)}{4} = -1 \times 60$$

$$\Rightarrow 20(x+2) - 12(x+1) - 15(x-3) = -60$$

$$\Rightarrow (20x + 40) - (12x + 12) - (15x - 45) = -60$$

$$\Rightarrow 20x + 40 - 12x - 12 - 15x + 45 = -60$$

$$\Rightarrow -7x + 73 = -60$$

$$\Rightarrow -7x = -60 - 73$$

$$\Rightarrow -7x = -133$$

$$\Rightarrow x = \frac{-133}{-7}$$

$$\Rightarrow x = 19$$

Hence, the value of x is 19.

Question 2(xi)

Solve:

$$\frac{3x-2}{3} + \frac{2x+3}{2} = x + \frac{7}{6}$$

Answer

$$\frac{3x-2}{3} + \frac{2x+3}{2} = x + \frac{7}{6}$$

Since L.C.M. of denominators 3 and 2 = 6, multiply each term with 6 to get:

$$\Rightarrow \frac{6(3x-2)}{3} + \frac{6(2x+3)}{2} = x \times 6 + \frac{7 \times 6}{6}$$

$$\Rightarrow 2(3x - 2) + 3(2x + 3) = 6x + 7 \times 1$$

$$\Rightarrow (6x - 4) + (6x + 9) = 6x + 7$$

$$\Rightarrow 6x - 4 + 6x + 9 = 6x + 7$$

$$\Rightarrow 12x + 5 = 6x + 7$$

$$\Rightarrow 12x - 6x = 7 - 5$$

$$\Rightarrow 6x = 2$$

$$\Rightarrow x = \frac{2}{6}$$

$$\Rightarrow x = \frac{1}{3}$$

Hence, the value of x is $\frac{1}{3}$.

Question 2(xii)

Solve:

$$x - \frac{x-1}{2} = 1 - \frac{x-2}{3}$$

Answer

$$x - \frac{x-1}{2} = 1 - \frac{x-2}{3}$$

$$\Rightarrow \frac{x}{1} - \frac{x-1}{2} = 1 - \frac{x-2}{3}$$

Since L.C.M. of denominators 1, 2 and 3 = 6, multiply each term with 6 to get:

$$\Rightarrow \frac{x \times 6}{1} - \frac{6(x-1)}{2} = 1 \times 6 - \frac{6(x-2)}{3}$$

$$\Rightarrow x \times 6 - 3(x-1) = 6 - 2(x-2)$$

$$\Rightarrow 6x - (3x-3) = 6 - (2x-4)$$

$$\Rightarrow 6x - 3x + 3 = 6 - 2x + 4$$

$$\Rightarrow 3x + 3 = 10 - 2x$$

$$\Rightarrow 3x + 2x = 10 - 3$$

$$\Rightarrow 5x = 7$$

$$\Rightarrow x = \frac{7}{5}$$

Hence, the value of x is $\frac{7}{5}$.

Question 2(xiii)

Solve:

$$\frac{9x+7}{2} - \left(x - \frac{x-2}{7}\right) = 36$$

Answer

$$\frac{9x+7}{2} - \left(x - \frac{x-2}{7}\right) = 36$$

$$\Rightarrow \frac{9x+7}{2} - x + \frac{x-2}{7} = 36$$

$$\Rightarrow \frac{9x+7}{2} - \frac{x}{1} + \frac{x-2}{7} = 36$$

Since L.C.M. of denominators 2 and 7 = 14, multiply each term with 14 to get:

$$\Rightarrow \frac{14(9x + 7)}{2} - \frac{x \times 14}{1} + \frac{14(x - 2)}{7} = 36 \times 14$$

$$\Rightarrow 7(9x + 7) - x \times 14 + 2(x - 2) = 504$$

$$\Rightarrow (63x + 49) - 14x + (2x - 4) = 504$$

$$\Rightarrow 63x + 49 - 14x + 2x - 4 = 504$$

$$\Rightarrow 51x + 45 = 504$$

$$\Rightarrow 51x = 504 - 45$$

$$\Rightarrow 51x = 459$$

$$\Rightarrow x = \frac{459}{51}$$

$$\Rightarrow x = 9$$

Hence, the value of x is 9.

Question 2(xiv)

Solve:

$$\frac{6x + 1}{2} + 1 = \frac{7x - 3}{3}$$

Answer

$$\frac{6x + 1}{2} + 1 = \frac{7x - 3}{3}$$

$$\Rightarrow \frac{6x + 1}{2} - \frac{7x - 3}{3} = -1$$

Since L.C.M. of denominators 2 and 3 = 6, multiply each term with 6 to get:

$$\Rightarrow \frac{6(6x + 1)}{2} - \frac{6(7x - 3)}{3} = -1 \times 6$$

$$\Rightarrow 3(6x + 1) - 2(7x - 3) = -6$$

$$\Rightarrow (18x + 3) - (14x - 6) = -6$$

$$\Rightarrow 18x + 3 - 14x + 6 = -6$$

$$\Rightarrow 4x + 9 = -6$$

$$\Rightarrow 4x = -6 - 9$$

$$\Rightarrow 4x = -15$$

$$\Rightarrow x = -\frac{15}{4}$$

Hence, the value of x is $-\frac{15}{4}$.

Question 3

After 12 years, I shall be 3 times as old as I was 4 years ago. Find my present age.

Answer

Let my present age be x years.

According to question, after 12 years, I shall be 3 times as old as I was 4 years ago.

$$\Rightarrow (\text{present age} + 12 \text{ years}) = 3(\text{present age} - 4 \text{ years})$$

$$\Rightarrow (x + 12) = 3(x - 4)$$

$$\Rightarrow x + 12 = 3x - 12$$

$$\Rightarrow 3x - x = 12 + 12$$

$$\Rightarrow 2x = 24$$

$$\Rightarrow x = \frac{24}{2}$$

$$\Rightarrow x = 12$$

Hence, my present age is 12 years.

Question 4

A man sold an article for ₹ 396 and gained 10% on it. Find the cost price of the article.

Answer

Given:

S.P. = ₹ 396

Profit = 10%

Let the C.P. of the article be ₹ x .

As we know,

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100$$

$$\Rightarrow 10 = \frac{\text{Profit}}{x} \times 100$$

$$\Rightarrow \text{Profit} = \frac{10 \times x}{100}$$

$$\Rightarrow \text{Profit} = \frac{10x}{100}$$

$$\Rightarrow \text{Profit} = \frac{x}{10}$$

And,

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\Rightarrow \frac{x}{10} = 396 - x$$

$$\Rightarrow \frac{x}{10} + \frac{x}{1} = 396$$

Since L.C.M. of 10 and 1 = 10, multiply each term with 10 to get:

$$\Rightarrow \frac{1x \times 10}{10} + \frac{x \times 10}{1} = 396 \times 10$$

$$\Rightarrow 1x + 10x = 3960$$

$$\Rightarrow 11x = 3960$$

$$\Rightarrow x = \frac{3960}{11}$$

$$\Rightarrow x = 360$$

Hence, cost price of the article is ₹ 360.

Question 5

The sum of two numbers is 4500. If 10% of one number is 12.5% of the other, find the numbers.

Answer

The sum of two numbers is 4500.

Let the number be x . The other number is $(4500 - x)$.

So, 10% of one number is 12.5% of the other number.

$$10\% \text{ of } x = 12.5\% \text{ of } (4500 - x)$$

$$\Rightarrow \frac{10}{100} \times x = \frac{12.5}{100} \times (4500 - x)$$

$$\Rightarrow \frac{x}{10} = \frac{125}{1000} \times (4500 - x)$$

$$\Rightarrow \frac{x}{10} = \frac{(4500 - x)}{8}$$

By cross multiplying, we get :

$$\Rightarrow 8x = 10(4500 - x)$$

$$\Rightarrow 8x = 45000 - 10x$$

$$\Rightarrow 8x + 10x = 45000$$

$$\Rightarrow 18x = 45000$$

$$\Rightarrow x = \frac{45000}{18}$$

$$\Rightarrow x = 2500$$

$$\text{Other number} = (4500 - x)$$

$$= (4500 - 2500)$$

$$= 2000$$

Hence, the two numbers are 2500 and 2000.

Question 6

The sum of two numbers is 405 and their ratio is 8 : 7. Find the numbers.

Answer

The ratio of two numbers is 8 : 7.

Let the numbers be $8x$ and $7x$.

The sum of numbers is 405.

$$\Rightarrow 8x + 7x = 405$$

$$\Rightarrow 15x = 405$$

$$\Rightarrow x = \frac{405}{15}$$

$$\Rightarrow x = 27$$

$$\text{Numbers} = 8x = 27 \times 8$$

$$= 216$$

$$\text{And, } 7x = 27 \times 7$$

$$= 189$$

Hence, the numbers are 216 and 189.

Question 7

The ages of A and B are in the ratio 7 : 5. Ten years hence, the ratio of their ages will be 9 : 7. Find their present ages.

Answer

Let the present age of A be 7x years and the present age of B be 5x years.

According to the question:

Ten years hence, the ratio of their ages will be 9 : 7.

$$\Rightarrow \frac{\text{Present age of A} + 10}{\text{Present age of B} + 10} = \frac{9}{7}$$

$$\Rightarrow \frac{7x + 10}{5x + 10} = \frac{9}{7}$$

By cross multiplying, we get :

$$\Rightarrow 7(7x + 10) = 9(5x + 10)$$

$$\Rightarrow 49x + 70 = 45x + 90$$

$$\Rightarrow 49x - 45x = 90 - 70$$

$$\Rightarrow 4x = 20$$

$$\Rightarrow x = \frac{20}{4}$$

$$\Rightarrow x = 5$$

So, present age of A = 7x

$$= 7 \times 5$$

$$= 35 \text{ years}$$

And, present age of B = $5x$

$$= 5 \times 5$$

$$= 25 \text{ years}$$

Hence, the present ages of A and B are 35 years and 25 years, respectively.

Question 8

Find the number whose double is 45 greater than its half.

Answer

Let the number be x .

Double of the number is 45 greater than its half.

$$\Rightarrow 2x = 45 + \frac{1}{2}x$$

$$\Rightarrow 2x - \frac{1}{2}x = 45$$

$$\Rightarrow \frac{2x}{1} - \frac{x}{2} = 45$$

Since L.C.M. of denominators 1 and 2 = 2, multiply each term with 2 to get:

$$\Rightarrow \frac{2x \times 2}{1} - \frac{x \times 2}{2} = 45 \times 2$$

$$\Rightarrow 2x \times 2 - x = 90$$

$$\Rightarrow 4x - x = 90$$

$$\Rightarrow 3x = 90$$

$$\Rightarrow x = \frac{90}{3}$$

$$\Rightarrow x = 30$$

Hence, the number is 30.

Question 9

The difference between the squares of two consecutive numbers is 31. Find the numbers.

Answer

Let the two consecutive numbers be x , $(x + 1)$.

According to the question, the difference between the squares of two consecutive numbers is 31.

$$\Rightarrow (x + 1)^2 - x^2 = 31$$

Using the formula,

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

$$\Rightarrow x^2 + 1^2 + 2 \times x \times 1 - x^2 = 31$$

$$\Rightarrow x^2 + 1 + 2x - x^2 = 31$$

$$\Rightarrow 1 + 2x = 31$$

$$\Rightarrow 2x = 31 - 1$$

$$\Rightarrow 2x = 30$$

$$\Rightarrow x = \frac{30}{2}$$

$$\Rightarrow x = 15$$

$$\text{Other number} = (x + 1)$$

$$= 15 + 1$$

$$= 16$$

Hence, the two consecutive numbers are 15 and 16.

Question 10

Find a number such that when 5 is subtracted from 5 times the number, the result is 4 more than twice the number.

Answer

Let the number be x .

According to the question,

when 5 is subtracted from 5 times the number, the result is 4 more than twice the number

$$\Rightarrow 5x - 5 = 2x + 4$$

$$\Rightarrow 5x - 2x = 4 + 5$$

$$\Rightarrow 3x = 9$$

$$\Rightarrow x = \frac{9}{3}$$

$$\Rightarrow x = 3$$

Hence, the number is 3.

Question 11

The numerator of a fraction is 5 less than its denominator. If 3 is added to the numerator and denominator both, the fraction becomes $\frac{4}{5}$. Find the original fraction.

Answer

Let the denominator of the fraction be x .

The numerator of a fraction is 5 less than its denominator

So, numerator = $x - 5$

According to the question,

when 3 is added to the numerator and denominator both, the fraction becomes

$$\frac{4}{5}$$

$$\Rightarrow \frac{\text{numerator} + 3}{\text{denominator} + 3} = \frac{4}{5}$$

$$\Rightarrow \frac{(x - 5) + 3}{x + 3} = \frac{4}{5}$$

$$\Rightarrow \frac{x - 5 + 3}{x + 3} = \frac{4}{5}$$

$$\Rightarrow \frac{x - 2}{x + 3} = \frac{4}{5}$$

By cross multiplying, we get :

$$\Rightarrow 5(x - 2) = 4(x + 3)$$

$$\Rightarrow 5x - 10 = 4x + 12$$

$$\Rightarrow 5x - 4x = 10 + 12$$

$$\Rightarrow x = 22$$

$$\text{Denominator} = 22$$

$$\text{Numerator} = (x - 5)$$

$$= 22 - 5$$

$$= 17$$

Hence, the fraction is $\frac{17}{22}$.

Question 12

The difference between two numbers is 36. The quotient when one number is divided by other is 4. Find the two numbers.

Answer

Let the number be x.

It is given that the difference between the numbers is 36. So, the other number is $x - 36$.

According to the question, the quotient when one number is divided by the other is 4.

$$\Rightarrow \frac{x}{x - 36} = 4$$

By cross multiplying, we get :

$$\Rightarrow x = 4(x - 36)$$

$$\Rightarrow x = 4x - 144$$

$$\Rightarrow 4x - x = 144$$

$$\Rightarrow 3x = 144$$

$$\Rightarrow x = \frac{144}{3}$$

$$\Rightarrow x = 48$$

Other number be $(x - 36)$

$$= (48 - 36)$$

$$= 12$$

Hence, the two numbers are 48 and 12.

Question 13

Five years ago, Mohit's age was thrice of Manish. 10 years later, Mohit will be twice as old as Manish. Find their present ages.

Answer

Let the age of Manish be x .

According to the question, five years ago, Mohit was three times as old as Manish.

$$\Rightarrow \text{Mohit's age} - 5 \text{ years} = 3 \times (\text{Manish's age} - 5 \text{ years})$$

$$\Rightarrow \text{Mohit's age} - 5 = 3(x - 5)$$

$$\Rightarrow \text{Mohit's age} - 5 = 3x - 15$$

$$\Rightarrow \text{Mohit's age} = 3x - 15 + 5$$

$$\Rightarrow \text{Mohit's age} = 3x - 10 \dots\dots\dots(1)$$

And,

Ten years later, Mohit will be twice as old as Manish.

$$\Rightarrow \text{Mohit's age} + 10 \text{ years} = 2 \times (\text{Manish's age} + 10 \text{ years})$$

$$\Rightarrow \text{Mohit's age} + 10 = 2(x + 10)$$

$$\Rightarrow \text{Mohit's age} + 10 = 2x + 20$$

$$\Rightarrow \text{Mohit's age} = 2x + 20 - 10$$

$$\Rightarrow \text{Mohit's age} = 2x + 10$$

From equation (1),

$$\Rightarrow 3x - 10 = 2x + 10$$

$$\Rightarrow 3x - 2x = 10 + 10$$

$$\Rightarrow x = 20$$

$$\text{Mohit's age} = 3x - 10$$

$$= 3 \times 20 - 10$$

$$= 60 - 10$$

$$= 50$$

Hence, the present age of Mohit is 50 years, and the present age of Manish is 20 years.

Question 14

The denominator of a fraction is 3 more its numerator. If the numerator is increased by 7 and the denominator is decreased by 2, the fraction is 2. Find the sum of the numerator and the denominator of the fraction.

Answer

Let the numerator of the fraction be x .

The denominator of a fraction is 3 more than its numerator.

So, denominator = $x + 3$

If the numerator is increased by 7 and the denominator is decreased by 2, the fraction is 2.

$$\Rightarrow \frac{\text{numerator} + 7}{\text{denominator} - 2} = 2$$

$$\Rightarrow \frac{x + 7}{(x + 3) - 2} = 2$$

$$\Rightarrow \frac{x+7}{x+3-2} = 2$$

$$\Rightarrow \frac{x+7}{x+1} = 2$$

By cross multiplying, we get :

$$\Rightarrow (x+7) = 2(x+1)$$

$$\Rightarrow x+7 = 2x+2$$

$$\Rightarrow 2x - x = 7 - 2$$

$$\Rightarrow x = 5$$

$$\text{Denominator} = (x+3)$$

$$= 5+3$$

$$= 8$$

The fraction is $\frac{5}{8}$.

The sum of the numerator and the denominator of the fraction = $5 + 8 = 13$.

Hence, the sum of the numerator and the denominator of the fraction is 13.

Question 15

The digits of a two-digit number differ by 3. If the digits are interchanged and the resulting number is added to the original number the result is 143. Find the original number.

Answer

Let the original number be $10x + y$.

As it is given that, the digits of a two-digit number differ by 3.

$$\Rightarrow x - y = 3$$

$$\Rightarrow x = 3 + y \dots\dots\dots(1)$$

If the digits are interchanged, the new number is $10y + x$

The sum of interchanged number and original number is 143.

$$\Rightarrow (10y + x) + (10x + y) = 143$$

$$\Rightarrow 10y + x + 10x + y = 143$$

$$\Rightarrow 11y + 11x = 143$$

$$\Rightarrow 11(y + x) = 143$$

$$\Rightarrow y + x = \frac{143}{11}$$

$$\Rightarrow y + x = 13$$

From equation (1),

$$\Rightarrow y + (3 + y) = 13$$

$$\Rightarrow y + 3 + y = 13$$

$$\Rightarrow 2y + 3 = 13$$

$$\Rightarrow 2y = 13 - 3$$

$$\Rightarrow 2y = 10$$

$$\Rightarrow y = \frac{10}{2}$$

$$\Rightarrow y = 5$$

$$x = y + 3$$

$$x = 5 + 3$$

$$x = 8$$

$$\text{Original number} = 10x + y$$

$$= 10 \times 8 + 5$$

$$= 80 + 5$$

$$= 85$$

Hence, the original number is 85.

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